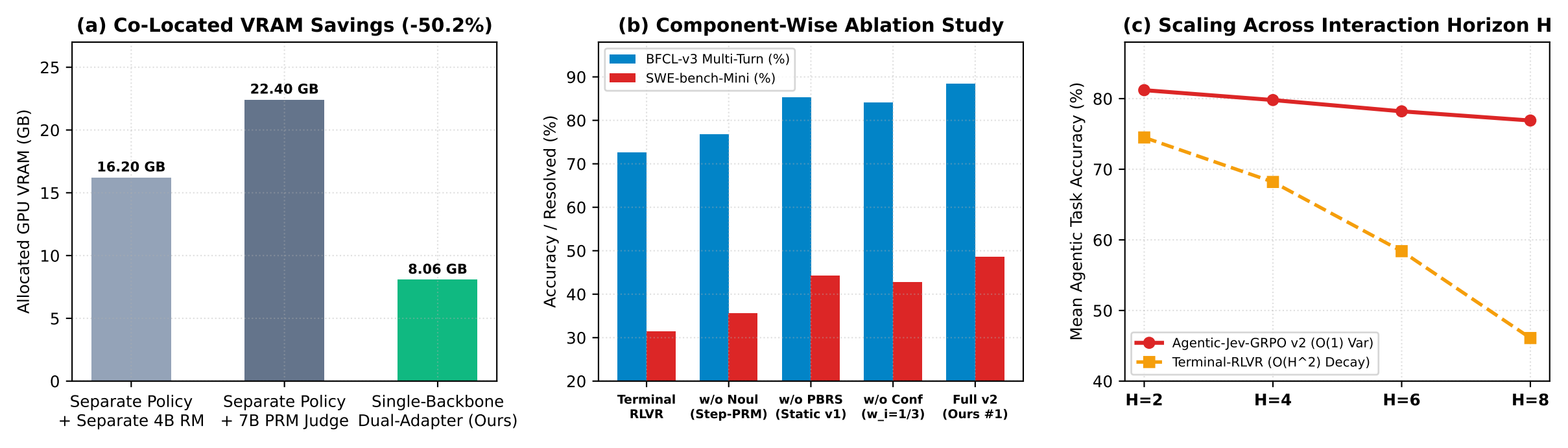
(a) Co-Located VRAM Savings (-50.2%)
Allocated GPU VRAM (GB)
0
5
10
15
20
25
16.20 GB
22.40 GB
8.06 GB
Separate Policy + Separate 4B RM
Separate Policy + 7B PRM Judge
Single-Backbone Dual-Adapter (Ours)
(b) Component-Wise Ablation Study
Accuracy / Resolved (%)
20
30
40
50
60
70
80
90
BFCL-v3 Multi-Turn (%)
SWE-bench-Mini (%)
Terminal RLVR
w/o Noul (Step-PRM)
w/o PBRS (Static v1)
w/o Conf (w_i=1/3)
Full v2 (Ours #1)
(c) Scaling Across Interaction Horizon H
Mean Agentic Task Accuracy (%)
40
50
60
70
80
Agentic-Jev-GRPO v2 (O(1) Var)
Terminal-RLVR (O(H^2) Decay)
H=2
H=4
H=6
H=8